

Model Paper - Database Management Systems

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Q1 (20 marks)

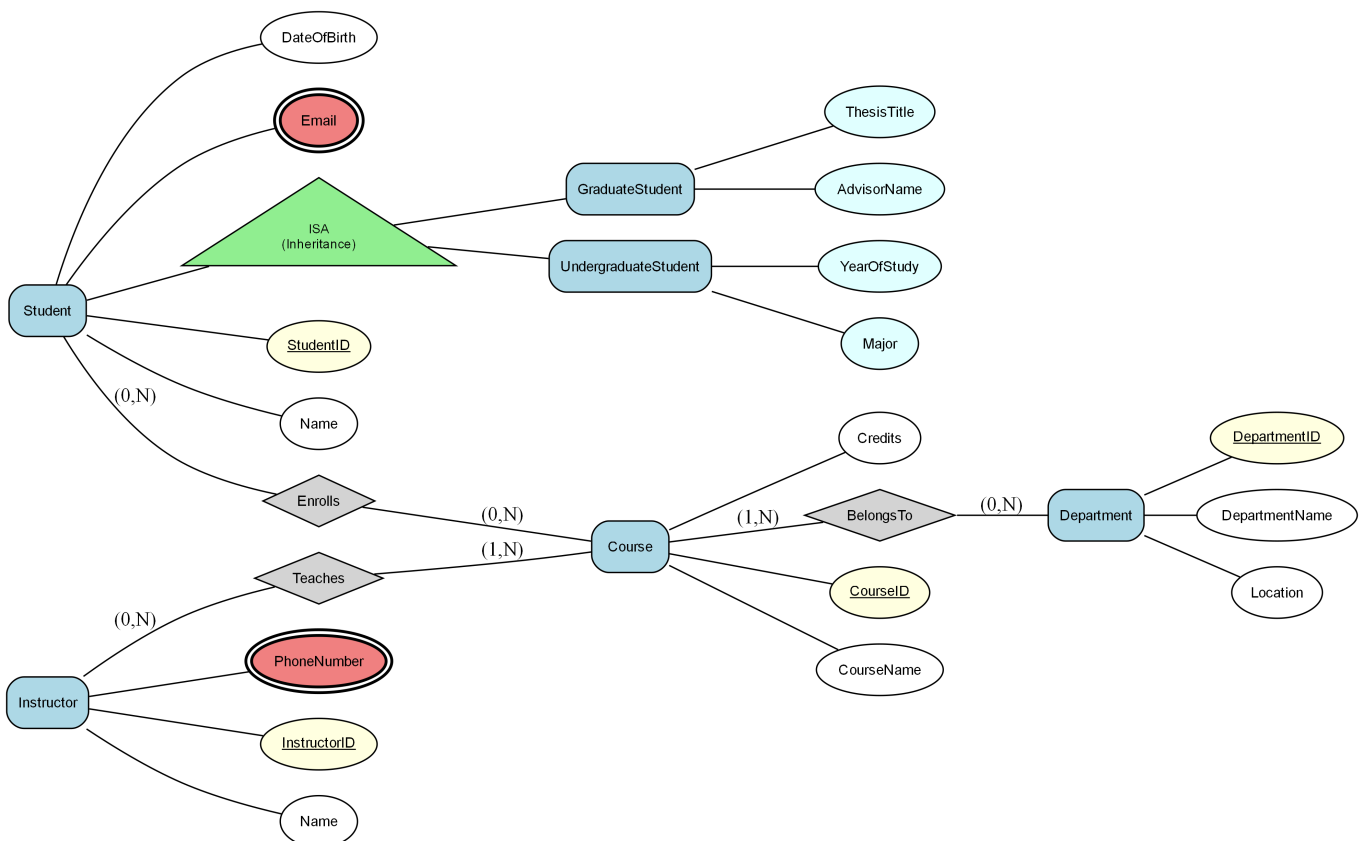
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university is managing its students, courses, and faculty using an enhanced entity-relationship model. The Student entity has attributes: StudentID, Name, Address, and PhoneNumbers. Faculty entity consists of attributes: FacultyID, Name, Department, and OfficeNumber. The Course entity includes attributes: CourseID, Title, Credits, and Schedule. The university uses an Enrolls relationship to link Students and Courses, which has the attribute EnrollmentDate. Additionally, there is a specialization in the Faculty entity, with a subtype representing ResearchFaculty which adds attributes such as ResearchArea and FundingAmount, while another subtype represents TeachingFaculty, adding attributes like TeachingLoad and CoursesTaught

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R :
 $F = \{AB \rightarrow C, B \rightarrow D, C \rightarrow AE, D \rightarrow B\}$.

- a) Find all the keys in relation R using the attribute closure method. (4 marks)
- b) Is R in 3NF? Give reasons for your conclusion. (3 marks)
- c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A university is designing a relational database to manage student enrollment and course offerings. The database schema includes relations for Students, Courses, and Enrollments. Each Student has a unique ID, a name, and email. Each Course is identified by a unique Course ID, a title, and a description. Enrollments track which Students are enrolled in which Courses. The database design must ensure data integrity and manage relationships effectively.

- a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)
- b) Code Segment: ````java String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql); pstmt.setString(1, "IT"); ResultSet rs =
pstmt.executeQuery(); ```` Which type of JDBC statement is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)
- c) Briefly explain how primary keys and foreign keys help maintain data integrity in this scenario. (2 marks)
- d) Briefly explain the steps needed to enforce referential integrity between the Students and Enrollments relations. (3 marks)
- e) A university is establishing role-based access control for its database system that manages Courses, Students, and Enrollments. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the Enrollments database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating student enrollment data.**
- i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)

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Q4 (40 marks)

Consider the following schema of a database designed for a Hospital: Patient (patientId: int, name: varchar(100), address: varchar(150), dob: date) Doctor (doctorId: int, name: varchar(100), specialty: varchar(50), phone: varchar(15)) Appointment (appointmentId: int, appointmentDate: date, patientId: int, doctorId: int, notes: varchar(200)) Prescription (prescriptionId: int, date: date, patientId: int, doctorId: int) The Patient table stores information about patients including their names, addresses, and dates of birth. The Doctor table holds details regarding doctors, their specialties, and contact numbers. The Appointment table manages appointment details, associating patients with doctors. The Prescription table maintains records of prescriptions linked to patients and the doctors who issued them. The 'Patient' table stores information about patients, including unique patient ID, name, date of birth, and contact details. The 'Doctor' table holds details about doctors, including unique doctor ID, name, specialization, and contact information. The 'Appointment' table manages appointment records, tracking scheduled appointments between patients and doctors with dates and times. The 'Prescription' table contains relevant information for the database system. Fee (feeId: int, patientId: int, amount: real, paymentStatus: varchar(20)) The 'Fee' table stores fee/payment details including payable amount and payment status.

a) Write SQL Queries to perform the following: i. Find the name and address of the patient who has an appointment on '2023-11-01'. ii. Find the names of all doctors who have a specialty in 'Cardiology'. iii. Find the appointment date and notes for all appointments of the patient named 'John Doe'.

i. Find the name and address of the patient who has an appointment on '2023-11-01'. (4 marks)

ii. Find

b) b) Create a function to calculate the total fee amount for a given patient. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each patient. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by patient. The patient pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)